Grade 8
Unit 10
Lesson 3 Practice

Name Answer Key
Date
Period

Determine which relations are functions. Then, explain your reasoning.

1. $\{(2,5),(7,5),(10,5),(15,5),(20,5)\}$

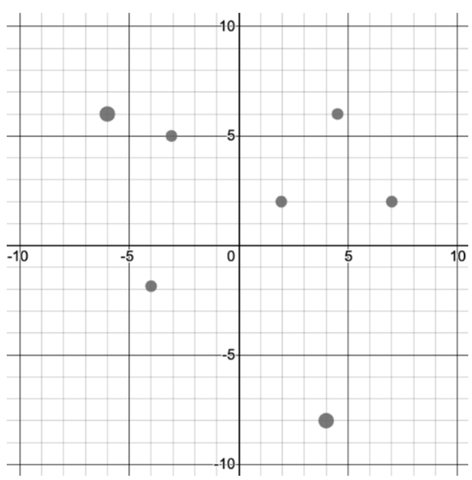
Function, the input values do not repeat.

2.

Input	Output
-5	-5
-5	0
-5	2
-5	5

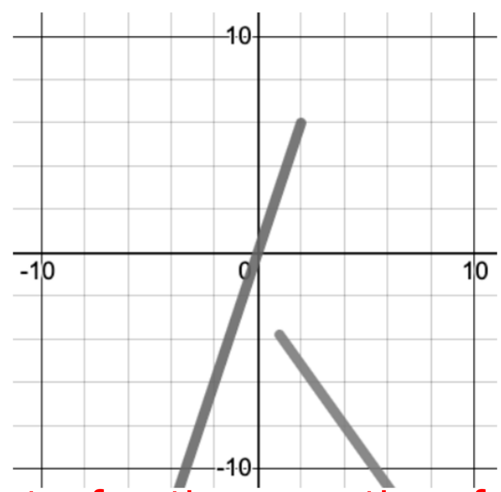
Not a function, the input value of -5 has four different outputs.

3.



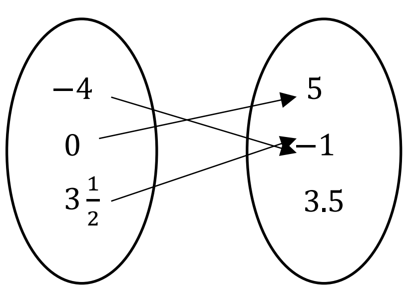
Function, each input has exactly one output.

4.



Not a function, a section of the graph has two outputs.

5. **Input** **Output**



Function, the input values do not repeat and match to one output.

6. $\{(0,-2),(-3,-7),(-5,10.2),(-3,0),(4.3,10)\}$

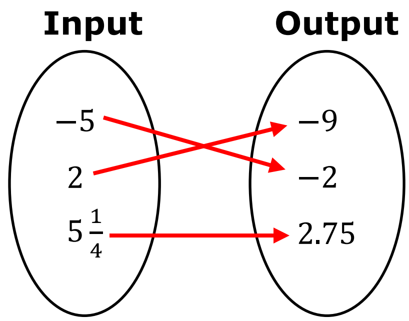
Not a function, the input value of -3 has two different output values.

7. Fill in the missing box so that the table does NOT represent a function:

Answers may vary: $-3, -1, 3$ or 5

Input	Output
-3	-4
-1	-4
-1	4
3	8
5	12

8. Fill in the lines on the mapping diagram so that it represents a function.

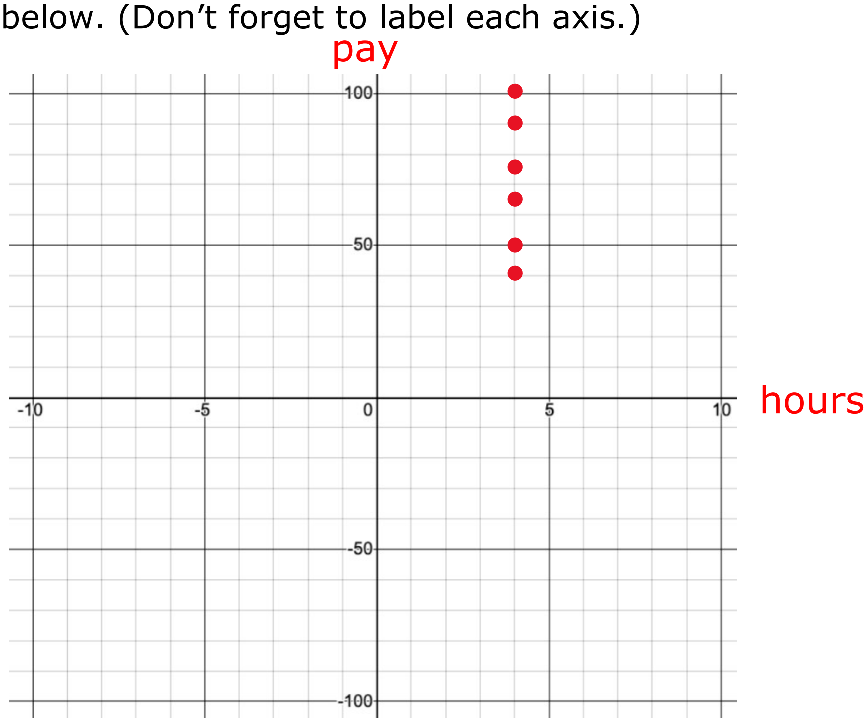


Answers may vary.

Everyone in the company only worked a half-day last Friday. Each person only worked for 4 hours and got paid their hourly wage. The table represents the pay, y, in dollars and the hours they worked, x.

x **(hours)**	4	4	4	4	4	4
y **(pay)**	40	50	65	75	90	100

9. Graph the relation below. (Don't forget to label each axis.)



10. Use the definition of a function to justify that the relationship between pay, y, and the hours worked, y, is NOT a function.

The relationship is not a function because the x-coordinate of 4 is paired with 6 different y-coordinates.